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DESIGN AND DEVELOPMENT OF AN ECOMMERCE PLATFORM FOR MOTORBIKE SPARE PARTS

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Date: …………………………….

A research proposal for an ecommerce platform for motorbike spare parts.

A Final Year Project Proposal submitted to the Department of Entrepreneurship Technology Leadership and Management
in partial fulfillment of the requirements for the award of a
Bachelor of Science Degree in Business Information Technology

Declaration

This project proposal is my original work, except where due acknowledgement is made in the text, and to the best of my knowledge has not been previously submitted to Jomo Kenyatta University of Agriculture and Technology or any other institution for the Award of a degree or diploma.

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SUPERVISOR CONFIRMATION:

This project proposal has been submitted to the Department of Entrepreneurship Technology Leadership and Management, Jomo Kenyatta University of Agriculture and Technology, with my approval as the University supervisor:

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ABSTRACT

In our current life motorbikes have had a rapid growth in the transportation sector, offering fast services to the people and just like any other machines they are prone to get maintenance and repairs done. Despite availability of so many shops that sell spare parts, riders often face challenges such as limited quality access of parts, high prices and long travel time and frustrations looking for the physical shops. E commerce stores remain undeveloped especially those that connect riders directly to vendors like Jumia connects buyers to different vendors on one platform.

This project aims to bridge this gap by providing an online marketplace tailored for motorbike riders and spare parts sellers. It will address the problem of inefficient access to affordable and quality motorbike spare parts by creating a platform that will enable riders to conveniently browse, order, and receive spare parts from local sellers.

The objective of the project was to develop a fully functional e-commerce application that connects motorbike riders with spare parts sellers, facilitating easy browsing, ordering, and payment.

The methodology employed involved the use of React for the frontend to ensure a responsive user interface, NodeJS for backend API development, and SQL for secure and organized data storage. User requirements were gathered through observation and surveys with motorbike riders and spare parts sellers to design a system that meets their specific needs.

The results showed that the system will facilitate efficient communication between sellers and riders. The platform successfully allowed users to register, browse available spare parts, place orders, make payments, track deliveries, and provide feedback.

In conclusion the platform provides a scalable and practical model that could be expanded to other regions or vehicle types in the future.

Contents

Chapter one

Introduction

1.1 Background Information

Over the last decade the motorbike sector has experienced exponential growth, driven by increasing urbanization, limited public transportation infrastructure, and the need for quick and reliable personal transport, serving as a key mode of transport for millions in both urban and rural areas. As a result, there has been a rampant need to offer maintenance and repair to the motorbikes. Riders often face challenges such as high costs, poor quality of parts, limited availability, and long travel distances to access reliable sellers (Kimani and Wicliff 2025).
In Kenya spare parts acquisition has remained as a challenge with riders having to visiting multiple shops to find quality components. This results in inefficiencies, higher operational costs, and time wastage for riders whose livelihoods depend on their motorbikes. The lack of standardized pricing and quality assurance mechanisms exposes riders to the risk of counterfeit or substandard parts, impacting safety and productivity (Kelvin, 2025). The gap in effective access to motorbike spare parts presents a pressing need for innovative solutions that leverage digital technologies to enhance efficiency and accessibility.

E-commerce platforms, which provide simplicity, transparency, and scalability, have completely changed how people may access goods and services. Platforms such as Jumia offer features such as online product catalogs, shopping carts, and secure payments however they have too generalized items and do not fully cater for the local motorbike rider who would want a store that fully deals with spare parts alone. The critical gaps that we are trying to solve are the absence of real-time delivery tracking, inadequate communication routes between riders and sellers, limited integration with local payment options like M-Pesa, and mobile-responsive interfaces tailored for low-bandwidth

The proposed project aims to bridge these gaps by developing a dedicated e-commerce platform for motorbike riders that connects them directly with spare parts sellers. Using React for the frontend, the system will provide a responsive and user-friendly interface accessible via smartphones, which are the primary access devices for most riders. NodeJS (Express) will handle backend processing, ensuring scalable and efficient handling of user requests, while SQL databases will securely manage transactional and inventory data. Key features will include product browsing, real-time order tracking, multiple payment options including M-Pesa or cash, seller ratings, and feedback systems to foster trust and transparency.

This study lays the groundwork for a workable and scalable digital intervention that tackles an actual issue in Kenya's transport sector by carefully examining market trends, rider requirements, and current e-commerce solutions.

1.1.1 System Process Overview

The system payment process is designed to ensure secure, fast, and transparent transactions between motorbike riders and spare parts sellers using M-Pesa integration.

Step 1: User Authentication

To begin any payment, the user needs to be registered and logged into the system. This guarantees that every transaction is associated with authenticated user accounts.

Step 2: Order Confirmation

The rider checks out after choosing the necessary motorbike spare parts. An order summary with item details, the total cost, and the chosen payment method (M-Pesa) is shown by the system.

Step 3: M-Pesa Payment Request (STK Push)

An M-Pesa STK Push request is sent to the rider's registered mobile phone number by the system when the rider confirms the order. The prompt asks the rider to input their M-Pesa PIN in order to authorize the payment.

Step 4: Payment Authorization by User

The rider reviews the payment details on their phone and enters the M-Pesa PIN to authorize the transaction. This step ensures user consent and transaction security.

Step 5: Payment Verification

After authorization, the system communicates with the M-Pesa API to verify the payment status. The transaction is validated as either successful or failed.

Step 6: Transaction Recording

If the payment is successful, the system automatically records the transaction details, including transaction ID, amount paid, date, and user information, in the SQL database.

Step 7: Order Status Update

The system updates the order status to “Paid” and notifies both the rider and the seller. If the payment fails, the system notifies the rider and allows them to retry the payment.

Step 8: Receipt Generation

A digital receipt is generated and made available to the rider within the system, providing proof of payment and enhancing transparency and accountability.

1.2 Statement of the problem

Even though Kenya's motorbike sector is expanding quickly, riders still struggle to find dependable and reasonably priced motorcycle spare parts. Many riders spend a lot of time driving to several physical stores in quest of parts, frequently coming across expensive, insufficient, or unavailable parts (Kimani and Wicliff 2025). This inefficiency not only increases operational costs but also affects riders’ productivity and safety, as delays or use of substandard parts can lead to breakdowns or accidents (Kenya National Boda Boda Association, 2021).

Existing e-commerce platforms such as Bodaboda360 (Classifieds for Bikes & Parts) have successfully digitized the sale of vehicle spare parts by providing product catalogs, shopping carts, and secure payment systems. However, it has existing gaps such as large non-standard stock parts are often posted individually rather than through a structured catalog, making search and comparison slow and no formal e-commerce workflow many transactions happen via direct contact without integrated checkout, delivery or tracking. (Mike,2025)

According to surveys conducted to (Mwangi & Otieno 2025), 40% of motorbike riders had bought inferior or fake parts in the previous year, and 65% of riders spend more than 30 minutes a day looking for spare parts. These numbers show how pervasive and serious the issue is.

The scope of the problem is focused on urban and peri-urban areas in Kenya where the demand for spare parts is highest and digital access is feasible. Adverse effects of not addressing this problem include reduced rider income due to downtime, increased operational costs, higher safety risks, and a fragmented spare parts supply chain. Features such as real-time product browsing, order placement, M-Pesa payment integration, delivery tracking, and a feedback mechanism for quality assurance directly address the inefficiencies identified. This approach not only improves access to quality parts but also creates a more organized and transparent marketplace for motorbike riders and spare parts sellers.

1.3 Proposed Solution

1. **Easy Ordering and Payment**
   Riders can order parts directly through the platform and pay safely using M-Pesa, Kenya's popular mobile payment system.
2. **Delivery and Tracking**
   Once an order is placed, riders can track it all the way until it reaches them, with estimated delivery times and notifications for updates.
3. **Organized Product Listings**
   All parts will be neatly categorized—like tires, brakes, or engines—with clear pictures, prices, and seller information. This makes it easy for riders to search, compare, and choose the right part.
4. **Feedback and Quality Checks**
   Riders can rate sellers and leave feedback on the parts they receive. This helps keep sellers honest and reduces the chances of buying fake or low-quality parts.

5. **Real-Time Stock Updates**

   Riders can see immediately if a part is available, so they don't have to visit multiple stores. They can also get notifications when parts are back in stock.

1.4 Objectives

1.4.1 Main Objectives

To create a specialized online market place that links Kenyan motorbike riders with motorbike spare parts vendors

1.4.2 Specific Objectives

1) **To design and implement a responsive user interface using React-** makes it simple for Kenyan motorbike riders and spare parts vendors to peruse merchandise, place orders, and communicate with the platform during the project's duration.
2) **To develop a secure backend system using NodeJS (Express)-** It oversees user accounts, orders, and inventory, guaranteeing dependable transaction processing and real-time data updates quantifiable through system testing.
3) **To integrate SQL database management-** to precisely track orders, payments, and inventory with quantifiable data integrity and dependability by storing and organizing product, user, and transaction data.
4) **To implement essential features such as M-Pesa payment integration, delivery tracking, and feedback mechanisms**- that improve user ease, transparency, and trust, as determined by functional testing and user input during the course of the project.

1.5 Scope of the Study

In this project, a web-based, mobile-responsive e-commerce platform for motorcycle spare parts in Kenya's urban and peri-urban areas is designed, developed, and evaluated. The system boundary is restricted to online activities such as product browsing, order placement, M-Pesa payment processing, delivery tracking, user registration, and feedback management for administrators, vendors, and riders. Only digital transactions and information management with React, NodeJS, and SQL are covered; manufacturing, physical storage, international transactions, and transportation logistics are not. The study is carried out during the project's duration to guarantee its viability and effective execution.

1.6 Assumptions

The participants in this study are assumed to possess smartphones, reliable internet access, and rudimentary digital abilities. Additionally, it is believed that consumers will be open to using the platform, give accurate information, and pay with M-Pesa. Furthermore, the analysis makes the assumption that mobile money services and technical infrastructure will continue to be dependable during the project's duration.

1.7 Limitations and Delimitations

Limited technical resources for system creation and maintenance, financial limitations, and perhaps inadequate internet access in some places are the study's key limitations. A few individuals can also be hesitant to switch from offline to online purchasing methods. Additionally, system performance may be impacted by security threats and system outages. These issues will be lessened by optimizing the system for low-bandwidth usage, providing basic user training, and securing it with data protection and secure authentication. It focuses solely on a web-based, mobile-responsive platform and is purposefully restricted to motorcycle replacement parts in the Kenyan market. It excludes other auto components, foreign markets, local mobile apps, and other payment methods, but it does cover M-Pesa and cash on delivery payments. These limits are in place to help manage resources, maintain concentration, and facilitate successful project completion within the allotted time.

1.8 System Research Questions

1. How does the proposed e-commerce system improve interaction and transactions between riders and vendors compared to existing systems?
2. What challenges do motorbike riders and spare parts vendors face when using traditional or existing e-commerce systems?
3. How does M-Pesa integration in the proposed system improve payment efficiency and security for both riders and vendors?

4. What challenges do spare parts vendors face when selling spare parts through existing systems?
5. How does order management and transaction records in the proposed system improve efficiency and accountability for vendors?

1.9 Problem Justification

Motorcycle spare parts are in great demand due to Kenya's motorbike industry's explosive expansion, yet riders continue to struggle to find dependable, reasonably priced, and high-quality parts. These difficulties include having to drive great distances to several stores, dealing with irregular pricing, and taking a chance on buying inferior or fake parts (Kelvin, 2022). In addition to raising riders' operating expenses, the current system's inefficiencies have an impact on their productivity, safety, and income. This initiative is driven by the pressing need to solve these inefficiencies by offering a digital solution that guarantees access to high-quality replacement parts and expedites the procurement process.

The situation is of significance because motorbike riders, who serve millions of people every day, are an essential component of Kenya's informal transportation network. Passenger safety may be compromised and riders' ability to make a living may be adversely impacted by delays in receiving replacement parts or the use of problematic components. By tackling this issue, the suggested initiative helps to improve riders' quality of life, boost the effectiveness of the spare parts supply chain, and assist regional companies by giving them access to a larger clientele.

2 CHAPTER TWO

Literature Review

2.1 Introduction

There is an urgent need for effective access to motorbike spare parts due to Kenya's expanding motorbike sector, but both riders and vendors have considerable difficulties in purchasing and selling these parts. The specific requirements of motorbike riders and spare parts suppliers are not entirely met by current e-commerce solutions, such as the Sharma Hawk platform, especially in areas like mobile optimization, M-Pesa integration, real-time order tracking, and direct buyer-seller interaction. This chapter highlights the features, benefits, and drawbacks of e-commerce systems, digital marketplaces, and online payment platforms like M-Pesa by reviewing pertinent literature.

2.2 Empirical Review

Empirical studies on Kenya's motorbike industry and ecommerce adoption reveal a strong need for a specialized platform connecting riders to spare part sellers. Research by NTSA shows that over 1.2 million riders rely heavily on spare parts for daily operations, yet surveys highlight persistent challenges such as counterfeit products, inconsistent pricing, and limited access to genuine suppliers. The suggested system is a specific online store for motorbike parts that caters to Kenyan motorbike riders. The system is assessed empirically by comparing it to marketplace-based systems like Shark, Jiji and current general purpose e-commerce platforms like Jumia and Kilimall. In order to demonstrate how the suggested system enhances current solutions by addressing industry-specific issues, the comparison focuses on system design, main functionality, user roles, and operational constraints. With features like role-based user access, model-specific spare part categorization, localized seller availability, and mobile payment integration tailored to the operational realities of motorbike riders, the proposed system is specifically designed to support motorcycle spare parts transactions, in contrast to the current platforms that are intended to support a wide range of products and users.

2.2.1 M-Pesa-Based Payment Integration in motorbike Spare Parts E-Commerce Platforms

In Kenya, e-commerce platforms for motorbike spare parts have increasingly adopted cashless payment systems through M-Pesa, providing customers with a seamless and secure transaction experience (KNBS, 2022). On selecting their preferred spare parts and proceeding to checkout, users are prompted to confirm payment via an M-Pesa STK push notification linked to their registered phone number. Upon successful completion of the transaction, the system automatically generates a digital receipt and updates the order status in real-time, ensuring transparency and traceability throughout the purchasing process. This integration not only reduces dependency on cash but also enhances trust and convenience for both sellers and buyers in the motorbike spare parts market.

2.2.2 System Architecture and Illustration

The motorbike spare parts e-commerce platform operates through several interconnected components:

1. **User Authentication Module** – Customers and sellers register and log in using verified credentials, ensuring secure access to the platform and protecting user data.
2. **Product Catalog and Ordering Engine** – This component allows users to browse, search, and select spare parts, while the system manages inventory levels and processes orders in real time.
3. **Payment Processing Module** – Payments are handled automatically through integrated M-Pesa transactions, triggering confirmations and digital receipts without manual intervention, reducing errors and improving transparency.

4. **Delivery Tracking and Management Module** – The system monitors order fulfillment and delivery status, providing real-time updates to customers and enabling sellers to manage dispatch efficiently.
5. **Data Management and Reporting Module** – A centralized database stores user profiles, product details, order histories, payment records, and delivery metrics, enabling platform administrators to analyze trends, monitor performance, and optimize operations.

2.3 Proposed System Architecture

The goal of the proposed motorcycle spare parts e-commerce system is to increase motorbike riders' accessibility, efficiency, and trust while buying motorbike spare parts. The system includes role-based user accounts for administrators, sellers, and riders; an integrated payment system (M-Pesa) for safe transactions; and real-time inventory management for precise product availability. This section describes the system's functional components and their primary functions. It also shows an example of the suggested system design, emphasizing how the layers work together to offer a smooth and dependable user experience.

2.3.1 Functional Components of the Proposed System

1. **E-Wallet / Integrated Payment System** – The system includes an integrated **M-Pesa wallet** feature, allowing riders to deposit funds or pay directly for spare parts. This reduces reliance on cash transactions, ensures instant payment confirmation, and provides riders with a simple way to manage their purchase budgets securely (World Bank, 2021a).
2. **Authentication Module** – Users are identified using unique **account credentials** (phone number or email) to ensure secure access. This module restricts actions based on roles (riders, sellers, administrators) and prevents unauthorized access to sensitive information or transactions.
3. **User Interface (Web & Mobile Application)** – Riders and sellers access a **responsive web and mobile interface**. Riders can browse products, check availability, place orders, and make payments, while sellers manage inventory, update prices, and track sales. The

interface is designed to be user-friendly, intuitive, and accessible from smartphones, tablets, or computers.

4. **Database Management System (DBMS)** – A centralized **SQL database** stores user profiles, product inventories, order history, transactions, and feedback. Real-time updates ensure accurate inventory tracking, transaction recording, and reliable reporting for system administrators (Brown & Smith, 2020).
5. **Order & Transaction Management** – The system tracks orders from placement to delivery, updating order status in real-time. Riders receive confirmation receipts with transaction IDs, while sellers can monitor order processing and manage dispatch, ensuring transparency and accountability in all transactions.
6. **Administrator Dashboard** – Administrators have a dedicated dashboard to manage users, approve sellers, monitor transactions, track inventory trends, and generate reports. This ensures that the system operates efficiently, helps prevent fraud, and provides insights for scaling the platform.

Illustration

Figure 2.3-1 Illustration of system functionalities.

2.4 System Design Requirements

For the successful implementation of the e-commerce motorbike platform**,** the following hardware and software components are required:

2.4.1 Hardware Requirements:

1. **Computer/ laptop –** it will be used for writing the codes and testing to ensure all the functionalities are working well that will be implemented to create the system, it therefore supports all software development activities.
2. **Server-** Acts as the central storage and processing unit for the system. It is responsible for processing user requests, handling authentication, managing product data, processing orders, and integrating with the M-Pesa payment gateway.

3. **Smartphones (For Riders and Vendors) -** Smartphones enable motorbikes riders and spare parts vendors to access the system through a web browser or mobile-friendly interface. Riders use smartphones to search for spare parts, place orders, and make M-Pesa payments, while vendors use them to manage products, confirm orders, and update order status.

2.4.2 Software Requirements

1. **Operating System (Windows, macOS, Linux)-** it provides the environment for development, testing, and server deployment. It ensures compatibility with development tools, databases, and the application runtime.
2. **Frontend Development Tools (React, Node Package Manager)** - React is used to build a responsive and user-friendly interface for riders and vendors. NPM manages dependencies and libraries required for running and maintaining the frontend.
3. **Backend Development Tools (Node.js and Express.js)** - Node.js executes server-side JavaScript, while Express.js handles APIs, HTTP requests, and application logic. Together, they manage user authentication, orders, and M-Pesa payment integration.
4. **Database Management System (SQL / MySQL)** - The SQL database stores user, vendor, product, and transaction data securely. It ensures fast data retrieval and maintains data integrity.
5. **Visual Studio Code-** The primary development environment used to write, debug, and test the system's front-end and back-end code
6. **Integrated Payment API (M-Pesa API)-** The M-Pesa API enables secure, real-time mobile money transactions within the platform. It facilitates instant payment verification and logs all transactions automatically.
7. **Web Browser -** Browsers are used for accessing and testing the e-commerce platform. Users rely on them to interact with the system on both computers and mobile devices.

3 CHAPTER THREE

SYSTEM DEVELOPMENT METHODOLOGIES

3.1 Introduction

System development methodologies are structured approaches used to plan, design, implement, and maintain software systems. These methodologies provide guidelines and frameworks to ensure efficient project management, timely delivery, and high-quality software development. Among the most widely used methodologies are Agile and Waterfall, each with its own advantages and limitations.

3.2 System Design Methodologies

The e-commerce platform's design and development are guided by both Waterfall and Agile approaches. The combination allows for flexibility and ongoing evolution while guaranteeing that the system is well-planned, structured, and documented. In accordance with the project's functional and non-functional requirements, the techniques facilitate the design of the system architecture, components, interfaces, and diagrams.

a) Waterfall Methodology

The Waterfall model is a linear and sequential approach to software development. Each phase: requirement Analysis, design, implementation, testing, deployment, and maintenance must be completed before moving to the next phase Sommerville, I. (2016). It is best suited for projects with well-defined requirements and minimal expected changes.

Advantage:

Clear Structure: The sequential nature of the Waterfall model ensures that each phase is clearly defined and properly documented.

Predictability: Project scope, timelines, and costs are determined early, making planning and management easier.

Well-Documented Process: Extensive documentation makes it suitable for academic projects and systems requiring formal approval.

Easy to Manage: Progress can be easily monitored since each phase has specific deliverables.

Best for Stable Projects: Works well when system requirements are fixed and unlikely to change significantly.

Disadvantage:

Inflexibility: Once a phase is completed, making changes becomes difficult and costly.

Late Testing: Errors may only be discovered during the testing phase, increasing correction costs.

Time-Consuming: The system is delivered only after all phases are completed.

Delayed Client Feedback: Users interact with the system only at the end of development.

High Risk if Requirements Are Incorrect: Mistakes made in early stages affect the entire system.

b) Agile Methodology

Agile is a flexible, iterative methodology that encourages frequent releases, ongoing teamwork, and client feedback. Teams can adjust to changes and improve the product depending on user needs because development can place in brief cycles known as sprints. Agile is frequently applied in fast-paced, dynamic settings Schwaber, K., & Sutherland, J. (2020)

Advantage:

Flexibility: System requirements and features can be modified at any stage of development.

Faster Delivery: Continuous releases ensure faster availability of system features.

Customer Involvement: Regular feedback improves system usability and relevance.

Early Detection of Issues: Continuous testing reduces bugs and system errors.

Improved Team Collaboration: Agile promotes communication and teamwork among developers.

Disadvantage:

Lack of Predictability: It is difficult to accurately estimate timelines and costs.

Requires Skilled Team: Agile development requires experienced developers and strong coordination.

Limited Documentation: Focus on functionality may result in minimal documentation.

Not Ideal for Fixed-Scope Projects: Less suitable where requirements must remain unchanged.

Risk of Scope Creep: Frequent changes can lead to uncontrolled expansion of project scope.

3.3 Fact Finding Techniques

Fact-finding techniques are used to gather accurate information and understand user needs, existing challenges, and system requirements for the proposed e-commerce platform. The following techniques were applied:

1. **Interviews-**

   They were conducted with motorbike riders and spare parts sellers to gather first-hand information about their experiences when purchasing or selling spare parts. This technique helped identify common challenges such as delays, lack of price transparency, and difficulties in accessing genuine spare parts.

   Advantage:

- **Rich Data Collection**: Interviews allow for in-depth and rich data collection. They provide an opportunity to gather detailed information, insights, and nuances that may not be captured through other data collection methods.
- **Personal Connection**: Interviews create a personal connection between the interviewer and the interviewee, which can foster trust and open communication. This can lead to more honest and candid responses.
- **Flexibility**: Interviews can be structured or unstructured, allowing for flexibility in the questioning approach. Researchers or interviewers can adapt their questions based on the interviewee's responses, enabling a deeper exploration of topics.
- **Clarification**: Interviews provide an opportunity to seek clarification on unclear or ambiguous responses. Follow-up questions can help ensure that the interviewer fully understands the interviewee's perspective.

Disadvantage:

limited Sample Size: Interviews usually involve few participants because they require significant time and effort, reducing the ability to generalize findings**.**

Social Desirability Bias: Respondents may give answers they think are acceptable rather than their true opinions or experiences.

Interviewer Effect: The interviewer's behavior, skills, or tone can influence how participants respond.

Time-Consuming: Interviews require substantial time for preparation, conducting sessions, and analyzing responses**.**

Resource Intensive: Conducting interviews can be costly due to the need for trained personnel and supporting resources.

2. **Questionnaires**

They were distributed to a larger group of potential users, including riders and sellers, to collect quantitative data on preferences such as payment methods, delivery expectations, and frequently purchased spare parts. This technique was effective in gathering data from many respondents within a short time and at a low cost.

Advantage:

Cost-Effective: Questionnaires are inexpensive to administer compared to interviews since they require fewer resources and personnel. They are suitable for collecting data from a large number of respondents.

Time-Efficient: Data can be collected within a short period, especially when questionnaires are distributed online. This allows faster analysis and decision-making.

Wide Coverage: Questionnaires enable data collection from a large and geographically dispersed population. This improves the representativeness of the findings.

Standardized Responses: All respondents answer the same set of questions, which improves consistency and simplifies data analysis. This reduces interviewer-related bias.

Disadvantage:

Low Response Rate: Some respondents may ignore or fail to complete the questionnaire, leading to incomplete data. This can affect the reliability of the results.

Limited Depth of Information: Questionnaires do not allow follow-up questions or clarification of responses. This limits the depth and detail of the information collected.

Misinterpretation of Questions: Respondents may misunderstand questions without the opportunity for clarification. This can result in inaccurate responses.

Dishonest Responses: Respondents may provide false or careless answers, especially when questionnaires are anonymous. This can reduce the validity of the data collected.

3. **Observation:**

It involved studying how motorbike riders currently purchase spare parts from physical shops and online platforms. This helped identify inefficiencies such as long waiting times, lack of stock visibility, and manual record-keeping. Observation provided a realistic understanding of existing processes and areas that require system automation.

Advantage:

Realistic Data Collection: Observation allows the researcher to study users in their natural environment without relying on self-reported information. This increases the accuracy and reliability of the data collected.

Captures Actual Behavior: It records what users actually do rather than what they say they do. This helps identify real processes, challenges, and inefficiencies.

Useful for Process Analysis: Observation is effective for understanding workflows and operational procedures. It helps identify gaps that may not be revealed through interviews or questionnaires.

Disadvantage:

Time-Intensive – Requires prolonged observation sessions.

Limited Scope – Observers can only monitor specific times and locations.

Privacy Concerns – Some users may feel uncomfortable being watched.

Observer Bias – The person observing may interpret behaviors subjectively.

No Direct Feedback – Observation does not provide reasons behind behaviors unless combined with other methods.

3.4 Feasibility Study for e-commerce platform for motorbike spare parts

3.4.1 Introduction

To ascertain whether the system can be effectively created and implemented, the research looks at the project from technical, economic, operational, and timetable feasibility viewpoints. To assess the potential and practicality of the suggested e-commerce platform that links motorbike riders with spare parts vendors, a feasibility study was carried out.

3.4.2 Feasibility

Technical Feasibility Study

In technical feasibility analysis, businesses assess whether the proposed endeavor is technically possible with the technology they already have or that already exists in the market. It scrutinizes the endeavor's technical requirements, constraints, and risks. By addressing concerns during the feasibility stage, the project ensures smooth deployment, optimal system performance, and long-term maintainability and integration with the hardware and software commonly used by motorbike riders and spare parts sellers.

Economic Feasibility Study

The goal of this economic feasibility study is to determine whether the proposed e-commerce platform for motorbike riders is financially viable. It evaluates the initial investment, ongoing operational costs, and potential financial benefits to ensure the platform is cost-effective for both riders and the business stakeholders. By conducting an economic feasibility analysis, it is ensured that the anticipated benefits such as better market access, less manual transaction handling, and enhanced sales efficiency justify the financial expenditure needed to construct and run the platform.

Legal and Regulatory Feasibility

Legal and regulatory viability evaluates the planned e-commerce platform's compliance with Kenyan laws, rules, and industry standards pertaining to digital transactions, online enterprises, and consumer protection. Key considerations include compliance with the Kenya Information and Communications Act, Data Protection Act 2019, consumer protection regulations, and digital payment guidelines for mobile money platforms such as M-Pesa. Ensuring adherence to these legal requirements protects user data, secures transactions, and provides a trustworthy platform for both riders and sellers.

Operational Feasibility

It evaluates whether the suggested e-commerce platform can be successfully put into place and utilized in the regular business operations of both spare parts vendors and motorbike riders. It assesses how well the system can satisfy user needs, increase productivity, and seamlessly integrate with current workflows. It ensures that the platform will be practical, convenient, and sustainable for all users, increasing adoption rates and supporting long-term operational success.

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